

SIMATS ENGINEERING

Department of Artificial Intelligence and Expert Systems

ASSESSMENT TOOL 2

Logic Reasoning & Mapping Exercise

Course Name: Artificial Intelligence and Expert Systems

Course Code: MLA01

Course Outcome: CO3 — Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning (BTL 3)

Weightage: 15% Total Marks: 100 Time: 60 Minutes

Question Attempted: Q.No 1 — Smart Healthcare Diagnosis

Student Details

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Code of Conduct

I S.Roshni (Reg No. 192525185) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Roshni

Q1 — Smart Healthcare Diagnosis: Full Solution

Scenario: A hospital AI system identifies possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.

(i) Problem Analysis and Knowledge Identification

Entities

- Ravi — the patient under diagnosis (constant)

Facts (given)

- Ravi has fever
- Ravi has cough
- Ravi has body pain

Predicates

- Fever(x) — x has fever
- Cough(x) — x has cough
- BodyPain(x) — x has body pain
- Flu(x) — x has flu
- ViralInfection(x) — x has a viral infection
- NeedsMedicalAttention(x) — x requires medical attention

Variables and Constants

- Variable: x (ranges over the set of patients)
- Constant: Ravi

Rules (given, in natural language)

- R1: If a patient has fever and cough, the patient may have flu.
- R2: If a patient has flu and body pain, the patient may have a viral infection.
- R3: If a patient has a viral infection, the patient requires medical attention.

(ii) Representation using Propositional Logic

Since propositional logic has no variables, each proposition is instantiated for Ravi:

$p = \text{Fever}(\text{Ravi})$ $q = \text{Cough}(\text{Ravi})$ $r = \text{BodyPain}(\text{Ravi})$
 $s = \text{Flu}(\text{Ravi})$ $t = \text{ViralInfection}(\text{Ravi})$ $u = \text{NeedsMedicalAttention}(\text{Ravi})$

Facts

p, q, r

Rules

R1: $(p \wedge q) \rightarrow s$

R2: $(s \wedge r) \rightarrow t$

R3: $t \rightarrow u$

(iii) Representation using First-Order Logic (FOL)

Using the universal quantifier $\forall$ (for all patients x), the rules generalise across every patient, while the facts about Ravi are ground atoms:

Facts: $\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \wedge \text{BodyPain}(\text{Ravi})$

R1: $\forall x \ (\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x))$

R2: $\forall x \ (\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x))$

R3: $\forall x \ (\text{ViralInfection}(x) \rightarrow \text{NeedsMedicalAttention}(x))$

Query (goal to prove):

$\exists x \ (x = \text{Ravi} \wedge \text{ViralInfection}(x)) \quad \equiv \quad \text{ViralInfection}(\text{Ravi}) ?$

Knowledge Mapping Diagram — Smart Healthcare Diagnosis (Ravi)

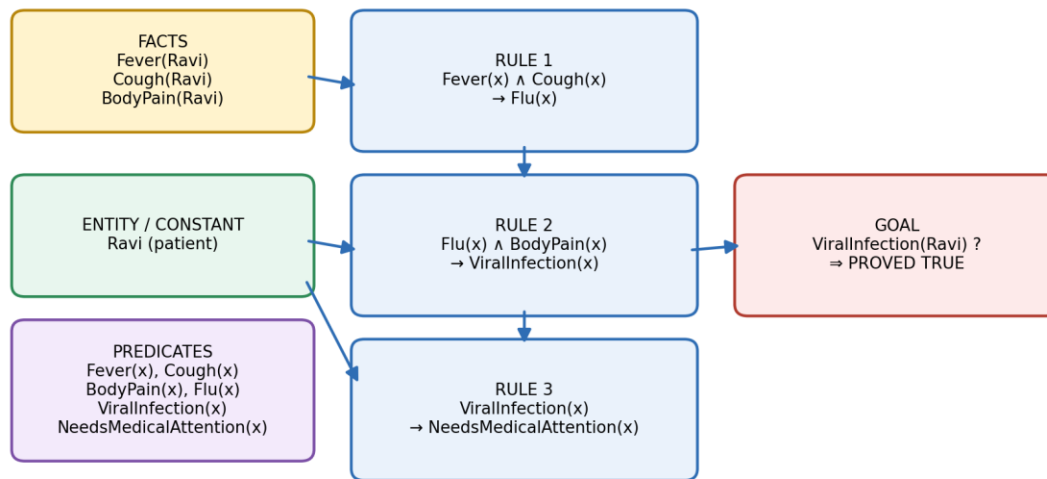


Fig. 1 — Knowledge mapping diagram: facts, entity, predicates and rules for Ravi's diagnosis

(iv) Unification and Substitution

To apply R1 to Ravi's facts, the variable x in the rule's antecedent must be unified with the constant Ravi appearing in the facts.

Unify $\text{Fever}(x)$ with $\text{Fever}(\text{Ravi})$

$\text{UNIFY}(\text{Fever}(x), \text{Fever}(\text{Ravi})) \rightarrow \text{Most General Unifier (MGU)} = \{ x / \text{Ravi} \}$

Check consistency with $\text{Cough}(x)$ and $\text{Cough}(\text{Ravi})$

$\text{UNIFY}(\text{Cough}(x), \text{Cough}(\text{Ravi})) \rightarrow \text{MGU} = \{ x / \text{Ravi} \}$ (consistent with the substitution above)

Since both unifications agree on the same substitution $\{x/\text{Ravi}\}$, this substitution is applied throughout R1, R2 and R3, converting the universally quantified rules into ground clauses specific to Ravi:

$R1[x/Ravi]: \text{Fever}(Ravi) \wedge \text{Cough}(Ravi) \rightarrow \text{Flu}(Ravi)$
 $R2[x/Ravi]: \text{Flu}(Ravi) \wedge \text{BodyPain}(Ravi) \rightarrow \text{ViralInfection}(Ravi)$
 $R3[x/Ravi]: \text{ViralInfection}(Ravi) \rightarrow \text{NeedsMedicalAttention}(Ravi)$

(v) Resolution Proof — Does Ravi have a Viral Infection?

Convert every rule and fact to clause (CNF) form, negate the goal, and resolve until the empty clause (NIL) is derived.

Clause set

$C1: \neg \text{Fever}(Ravi) \vee \neg \text{Cough}(Ravi) \vee \text{Flu}(Ravi)$ [from R1]
 $C2: \neg \text{Flu}(Ravi) \vee \neg \text{BodyPain}(Ravi) \vee \text{ViralInfection}(Ravi)$ [from R2]
 $C3: \neg \text{ViralInfection}(Ravi) \vee \text{NeedsMedicalAttention}(Ravi)$ [from R3]
 $F1: \text{Fever}(Ravi)$ $F2: \text{Cough}(Ravi)$ $F3: \text{BodyPain}(Ravi)$ [facts]
 $NG: \neg \text{ViralInfection}(Ravi)$ [negated goal]

Resolution steps

- Step 1: Resolve C1 with F1 ($\text{Fever}(Ravi)$) $\rightarrow R1': \neg \text{Cough}(Ravi) \vee \text{Flu}(Ravi)$
- Step 2: Resolve R1' with F2 ($\text{Cough}(Ravi)$) $\rightarrow R2': \text{Flu}(Ravi)$
- Step 3: Resolve C2 with R2' ($\text{Flu}(Ravi)$) $\rightarrow R3': \neg \text{BodyPain}(Ravi) \vee \text{ViralInfection}(Ravi)$
- Step 4: Resolve R3' with F3 ($\text{BodyPain}(Ravi)$) $\rightarrow R4': \text{ViralInfection}(Ravi)$
- Step 5: Resolve NG ($\neg \text{ViralInfection}(Ravi)$) with R4' ($\text{ViralInfection}(Ravi)$) $\rightarrow \text{NIL}$ (empty clause)

The derivation of NIL is a contradiction, which proves that the negated goal was false — hence the original goal $\text{ViralInfection}(Ravi)$ is TRUE.

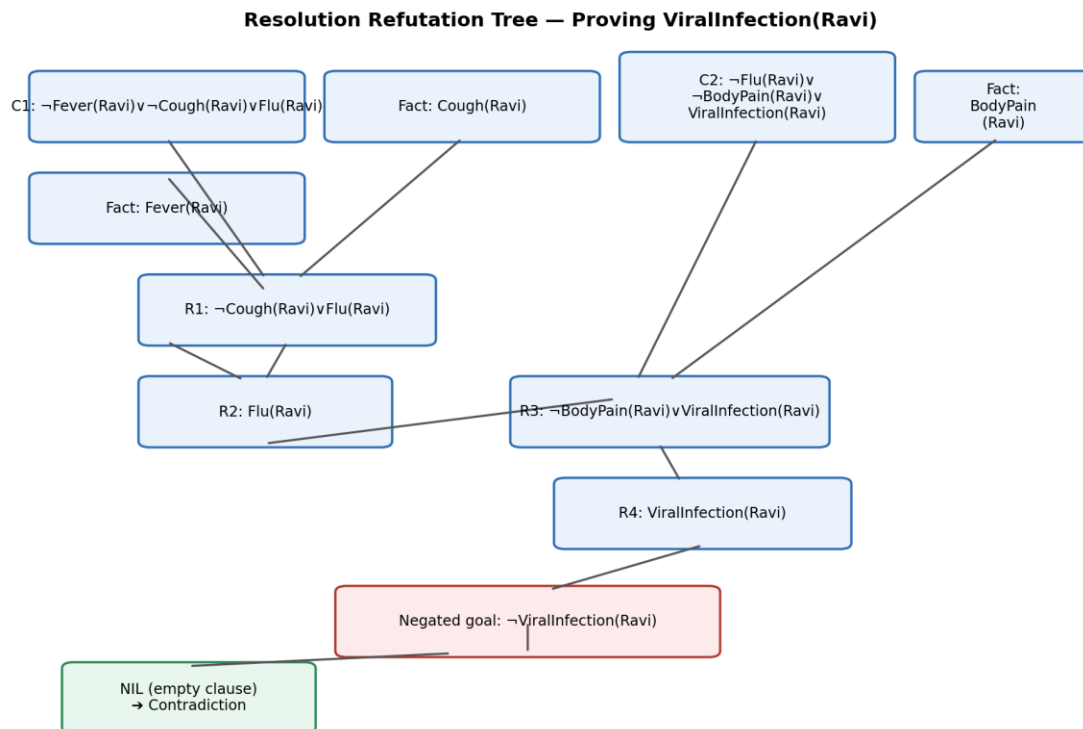


Fig. 2 — Resolution refutation tree proving $\text{ViralInfection}(Ravi)$

(vi) Complete Inference Sequence and Final Conclusion

- 1. Fever(Ravi) and Cough(Ravi) are known facts.
- 2. Applying R1 with {x/Ravi}: $\text{Fever(Ravi)} \wedge \text{Cough(Ravi)} \rightarrow \text{Flu(Ravi)} \Rightarrow \text{Flu(Ravi)}$ is derived.
- 3. BodyPain(Ravi) is a known fact.
- 4. Applying R2 with {x/Ravi}: $\text{Flu(Ravi)} \wedge \text{BodyPain(Ravi)} \rightarrow \text{ViralInfection(Ravi)} \Rightarrow \text{ViralInfection(Ravi)}$ is derived.
- 5. Applying R3 with {x/Ravi}: $\text{ViralInfection(Ravi)} \rightarrow \text{NeedsMedicalAttention(Ravi)} \Rightarrow \text{NeedsMedicalAttention(Ravi)}$ is derived.

Forward-Chaining Inference Sequence — Ravi's Diagnosis

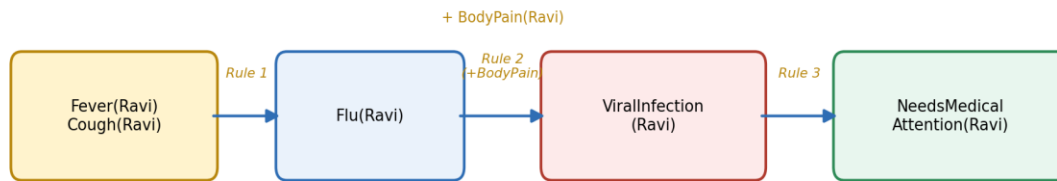


Fig. 3 — Forward-chaining inference sequence from symptoms to conclusion

Final Conclusion: Ravi has a viral infection [$\text{ViralInfection(Ravi)} = \text{TRUE}$] and therefore requires medical attention [$\text{NeedsMedicalAttention(Ravi)} = \text{TRUE}$].

Evaluation Rubrics

Criterion	Wt.	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to construct valid logical formulas.	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.
Total: 100%		Marks Obtained: _____ / 100			

Evaluator's Signature: _____ *Date:* _____